Abinash Sonowal

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Education

•Indian Institute of Technology Guwahati

B.Tech. in Computer Science and Engineering | Coursework: Computer Networks, Operating Systems Nov 2021 - May 2025

Experience

•Datametica Birds, Pune, India *Jun. 2025 - Present*

Member of Technical Staff - Product Team Pune

- **Pipeline reliability & debugging:** Hardened the batch pipeline that ingests, transforms and publishes client BI reports, raising end-to-end success from **84 to 487 of 672** by root-causing failures and fixing two core transformation modules. Built an end-to-end validation layer that certifies per-report coverage.
- **Ingestion-to-publishing service:** Built the Raven BI service end to end: a **Spring Boot REST API** (**12** endpoints for uploads, conversion runs, **Git publishing**, downloads and run history) on **PostgreSQL**, running conversions on a background executor, plus a **React**/**TypeScript** UI.
- **Transformation engines (OOP & design patterns):** Built Tableau and **Power BI→LookML** compilers in **Java**/**Groovy** (∼**17K** lines): **ANTLR** grammars with **Visitor**-based translation into **Apache Calcite** plans, extensible function registries, lineage in **Neo4j**, **150+ Spock tests**. Migrated **52** enterprise workbooks to **496** LookML artifacts.
- **Multithreaded agents:** Built **3 Kotlin**/**LangChain4j** agents (datasource parsing, **KPI** identification, SQL generation) that fan out batch work over a **thread pool** with **CompletableFuture**, behind a **Gradle CLI** emitting JSON.
- **AI code-review agent:** Built an **LLM**-based merge-request reviewer (**Python** CLI + **GitLab CI** job) on **OpenCode**: correctness and security/quality agents review each MR in a sparse **git worktree** at its head, and a lead agent dedupes and verifies their findings. Added **429** backoff with session resume, deadlines, token/cost logging and e-mailed reports; **200+ pytest** tests.
- **Automation tooling:** Built an assessment & volumetrics CLI that scans directories of BI workbooks and reports data connections and per-formula convertibility; prototyped in **Python**, then rebuilt in **Java**.

•Samsung R&D Institute -Bangalore(SRI-B) *May 2024 - Jul. 2024*

Research & Developer Intern - Visual Intelligence Team Bangalore

- Designed and optimized a lightweight **UNET** model in **PyTorch** for efficient on-device image deblurring, improving performance using skip connections and data augmentation with **Albumentations**.
- Worked with Samsung's ML deployment pipeline and successfully ported the optimized model to the **Samsung S24 Ultra S298** platform for real-time testing and validation.

Projects

•Whitejack — Multiplayer Card Games with Jev-Powered Bots *Sep. 2026 - Present*

Personal Project Live Demo | Github

- Built a browser-based multiplayer platform for **Poker**, **Hearts** and **Gin Rummy** with **Java 17**/**Spring Boot WebSockets** and a **React 19**/**TypeScript** client; deployed live with **Docker**.
- Designed a **concurrent, server-authoritative engine**: each room is an **actor** on its own thread with a bounded command queue for backpressure, a **ConcurrentHashMap** room registry and scheduled turn timers. Clients send only intents and get per-seat views with hidden cards redacted.
- Built AI bots as ordinary socket clients powered by TypeSafe's **Jev** decision model (via **OpenRouter**), chosen for speed and cost: each move ranks only the legal options for ∼**$0.00003** (∼700–800 input / 30–80 output tokens), with a heuristic fallback after a **3 s** timeout; reasoning is published only after each hand ends.
- Covered rules, redaction and end-to-end play with **169 tests** in **GitHub Actions** CI.

Technical Skills

- **Programming**: Java, Kotlin, Python, Groovy, C/C++, TypeScript, JavaScript
- **Core**: OOP, Design Patterns (Visitor, Registry, Actor), Multithreading & Concurrency (thread pools, CompletableFuture, blocking queues)
- **Backend & Infrastructure**: Linux, Spring Boot, REST APIs, HTTP/WebSockets, React, FastAPI, Docker, Gradle, GitLab CI, GitHub Actions
- **AI Engineering**: LangChain4j agents, OpenRouter, Claude Code (AI-assisted development)
- **Data & BI**: PostgreSQL, BigQuery, Neo4j, Teradata, Apache Calcite, ANTLR, Power BI (DAX, Power Query M), Tableau, Looker/LookML
- **Testing & Tools**: Spock, Git, GitLab, Jira, Vite

Achievements

•Codeforces Specialist, Maximum rating of 1563. *Codeforces Handle*

- Codeforces Round 968: Secured **Global Rank 272** among 34200 participants

•Rising Star Award, Datametica — employee recognition for impact and growth *2026*